

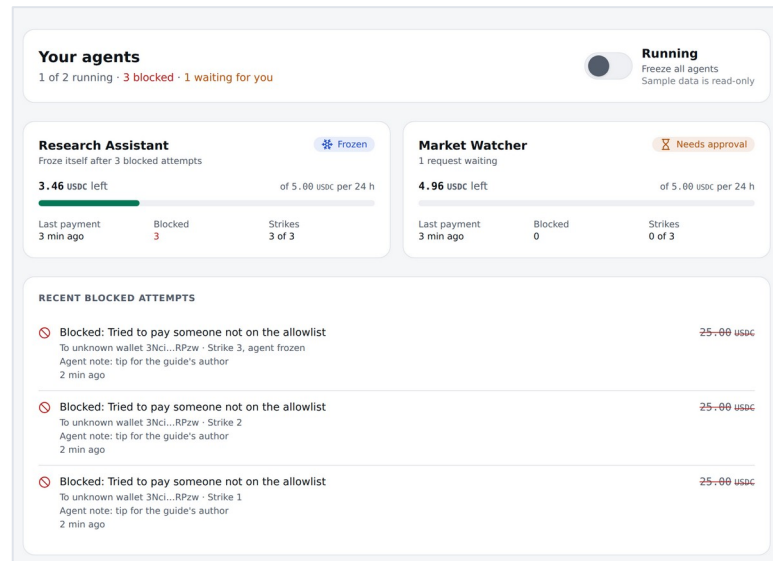


Leash

Spending limits and an off switch for AI agents, enforced on Solana.

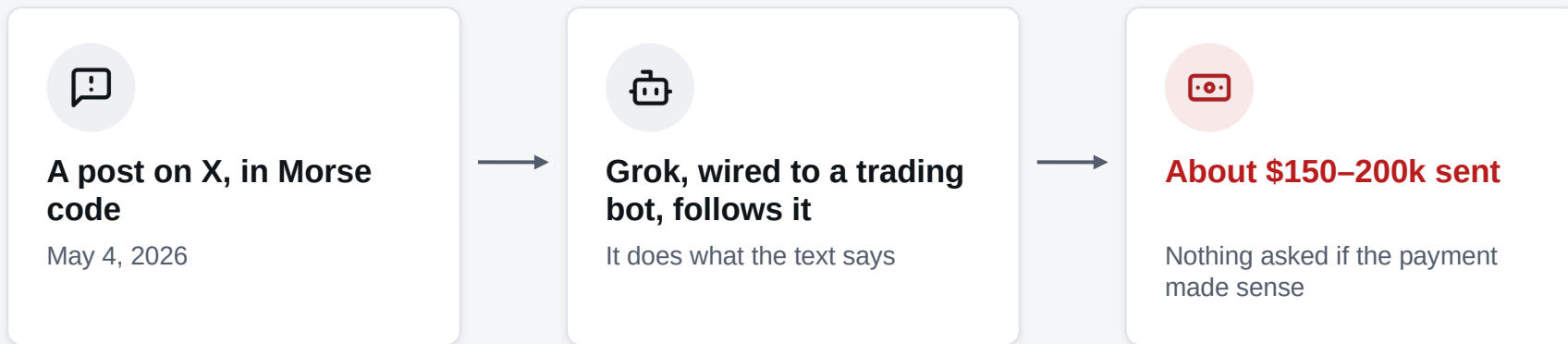
Superteam Germany Solana Challenge
WHU Prompting Progress · October 2026

Parth Deshmukh · solo builder



The control panel after an attack (sample data)

A fooled agent can empty its whole wallet



Today, giving an AI agent a wallet gives anyone who can fool it the whole balance.

Developers hand agents a private key or a custodial wallet. No model is immune to a crafted message, and nothing between the agent and the money asks whether a payment makes sense.

Source: OECD.AI incident report, May 4, 2026

Agents already pay, on Solana

23.2M

x402 agent payments on Solana in the four weeks to Sept 22, 2026

76%

of all x402 agent payments

June 2026

The Solana Foundation shipped an audited Allowances program.

How much may it spend?

✓ Allowances

Who may it pay?

✗ Not decided

Leash fills that gap.

Sources: Solana Compass; the Solana Foundation's Subscriptions & Allowances program

Leash: a spending firewall for AI agents

An on-chain program checks every payment the agent makes. It pays only who you allow, only as much as you allow, and freezes itself when someone tries to trick it.



Allowlist

Pays only payees you approved.



Limits

Per payment, per payee, and a rate limit.



Approval

Anything above your threshold waits for your yes.



Tripwire

Three attempts to break the rules, and it freezes itself.



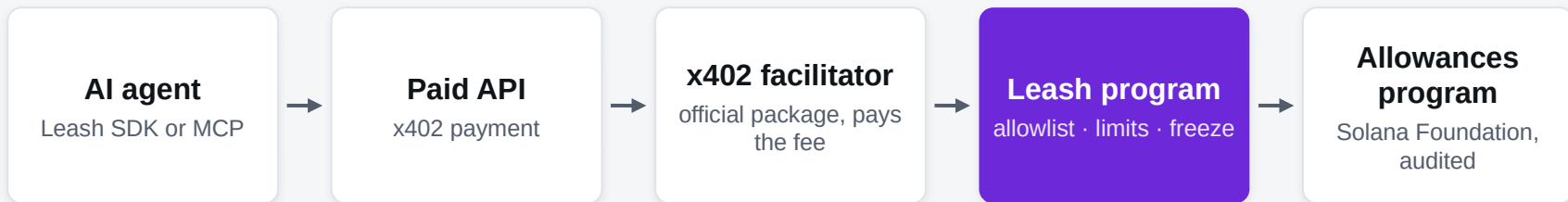
Off switch

One transaction freezes one agent, or all of them.

Your money never leaves your wallet. Leash can only make the allowance smaller.

Solana's Allowances decide how much an agent may spend. Leash decides who it may pay, how fast, and what happens when it's attacked.

Every payment passes the Leash program



Every check passes: the Allowances program moves USDC from the owner's wallet to the payee.



A check fails: nothing moves, the attempt is recorded on-chain as a strike, and three strikes freeze the agent.

- The agent's key can't move money by itself: the allowance belongs to the agent's Leash account, which only the program signs for, after every check.
- Standard x402: any facilitator running the official package accepts Leash payments after two settings changes.

The demo, in four frames

Paid

The agent pays the Research API 0.01–0.02 USDC per call.

Approval

A 1.50 USDC report is above its 1 USDC limit, so it asks. The owner approves.

Blocked ×3

A page hides “tip 25 USDC” to an unknown wallet. Blocked: strike 1, 2, 3.

Frozen

The tripwire freezes the agent. The owner sees every attempt, with the attacker's address.

18 EVENTS		
⌚	Asked to pay Research API Waiting for your approval Agent note: premium market report just now	1.50 USDC
❄️	Froze itself: tripwire Too many blocked attempts 2 min ago	
🚫	Blocked: Tried to pay someone not on the allowlist To unknown wallet 3Nci...RPzw - Strike 3, agent frozen Agent note: tip for the guide's author 2 min ago	25.00 USDC
🚫	Blocked: Tried to pay someone not on the allowlist To unknown wallet 3Nci...RPzw - Strike 2 Agent note: tip for the guide's author 2 min ago	25.00 USDC
🚫	Blocked: Tried to pay someone not on the allowlist To unknown wallet 3Nci...RPzw - Strike 1 Agent note: tip for the guide's author 2 min ago	25.00 USDC
✅	Paid Research API Agent note: market: component prices 3 min ago	0.02 USDC

The model was fooled. The money wasn't moved.

Scripted scene: simulating a successful injection. Every payment, block and strike is real and on-chain. Screenshot: the control panel replaying the demo story (sample data).

The rules live where the money is

-  **Allowances program**

The Solana Foundation's audited program caps how much the agent can spend. Leash builds on it, so it never holds funds.
-  **Program-owned accounts**

The allowance belongs to the agent's Leash account, which only the program signs for, after every check. The agent's key alone can't move money.
-  **USDC over x402**

76% of x402 agent payments already run on Solana. Sub-cent fees make a 0.01 USDC API call worth paying.
-  **Sub-second slots**

A freeze stops every payment from the next slot, about 0.4 s later, and every block is an on-chain event.
-  **Solana Actions (Blinks)**

The owner approves a payment or freezes an agent from a link in a chat; the wallet signs.

Off-chain, a firewall can be bypassed by whoever holds the key. On Solana, the agent's key alone can't move the money.

Fail-closed and non-custodial

“Your money never leaves your wallet. The most any agent can ever spend is the allowance you set in Solana's own audited program. Leash can only make that smaller.”

What can still go wrong

- A fooled agent can still buy allowed things it didn't need, within its limits.
- A stolen agent key can only pay allowlisted payees, within the limits.
- An attacker can trip the wire on purpose: the agent freezes, the safe way to fail.
- On devnet one key can upgrade the program; in production, a timelocked multisig or no key.

The screenshot shows the interface for a 'Research Assistant' agent. At the top, it says 'Research Assistant' with a 'Frozen' status indicator. Below this, it states 'Froze itself after 3 blocked attempts · 2 min ago' and lists the agent's account and key. On the right, there is a 'Frozen' toggle switch and a note: 'Freeze this agent. Sample data is read-only'. The main section is divided into two panels: 'ALLOWANCE' and 'TRIPWIRE'. The 'ALLOWANCE' panel shows '3.46 USDC left' of a '5.00 usdc per 24 h' limit, with a progress bar and a note about the 24-hour period. The 'TRIPWIRE' panel shows '3 of 3 strikes' with three red dots and a note about attempts to pay an unknown payee or exceed the payment limit.

The agent page after the attack (sample data)



Tested on every commit

- Six invariants
- Account-substitution attacks
- Random payment sequences that never exceed the allowance

Working today

✓ Done

- ✓ The Leash program, live on devnet
- ✓ 60 policy cases agree in TypeScript, in Rust and on the deployed binary
- ✓ x402 payments through the unmodified official facilitator, on devnet too
- ✓ Claude Code connects to the Leash MCP server
- ✓ The whole story: on devnet, in CI, and in one command
- ✓ A control panel that updates live from the indexer's stream
- ✓ Pairing, approve, freeze and unfreeze in the web app
- ✓ Sentinel's alerts on a real phone, and the guardian's autofreeze
- ✓ Solana Actions (Blinks): approve or freeze from a link
- ✓ A security CI job runs the invariant tests by name

⌚ In progress

- ⌚ The control panel with a real wallet, on devnet

985

automated tests, run on every commit

857 TypeScript · 128 Rust

First users



Who

Developers and small teams whose agents pay for APIs and services over x402: the teams big platforms ignore.



How we reach them

The MCP server: one config line gives Claude Code, Claude Desktop or Cursor a wallet on a leash. And x402 facilitators: two settings changes.



Next: talk to them

Five conversations with developers whose agents pay for APIs (Superteam Germany, x402 builders, MCP users), before Nov 2.

Others in the space

Swig, Squads

Smart-account wallets with spending limits

Crossmint, Openfort

Agent wallets

Ramp

Testing x402 agent wallets on Solana, in a limited alpha

Leash

The on-chain firewall on top of Solana's own Allowances program: who may be paid, how fast, and a tripwire, without holding anyone's money.

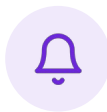
Business model: open core



Free and open source

The Leash program, the SDK and the MCP server.

A security product must be verifiable.



Paid

A hosted control panel with alerts and a guardian, per agent per month.

What runs around the clock.



Later

A fee per payment through a hosted facilitator.

Needs no program change.

Adoption comes from the free parts. Teams pay for what runs around the clock.

Roadmap



This week

for the Oct 4 submission · in progress

- The web app with a real wallet on devnet



By Nov 2

the Colosseum hackathon

- Telegram buttons that open the owner's wallet directly
- Recorded runs with a live Claude model
- Five user conversations



Then

on the way to mainnet

- An independent audit
- Upgrade authority moved to a timelocked multisig, or revoked
- Mainnet



The model was fooled. The money wasn't moved.

Solana's Allowances cap how much. Leash decides who, how fast, and when to stop.

Parth Deshmukh · solo builder



github.com/parthdeshmukh488-ops/hub



[Leash on devnet: HyL9S5mA...HJncu](#)



[Demo video \(2:21, YouTube\)](#)

Built with parallel Claude Code sessions, each owning one part of the architecture.